Functions: representations and reference functions

Lecture and worked examples

Function, image and preimage

A real-valued function associates to each real number x of its domain D_f a unique real number $f(x)$, called the image of x. A preimage of y is a real number $x \in D_f$ such that $f(x) = y$. We write $f : x \mapsto f(x)$. A number may have no preimage, one, or several.

Function, image and preimage

Let $g(x) = x^2 - 4x + 32$ on $\mathbb{R}$. Calculate $g(-5)$. Let $h(x) = 3x + 5$; determine the preimage of 6 under h.

Worked answer: Function, image and preimage

$g(-5) = 25 + 20 + 32 = 77$: 77 is the image of -5, and -5 is a preimage of 77. For h, $3x + 5 = 6 \iff x = 1/3$. The only preimage is $1/3$.

Function, image and preimage

For the square function, calculate the images of -3 and 3, then the preimages of 9, 0 and -1.

Worked answer: Function, image and preimage

$f(-3) = f(3) = 9$. The preimages of 9 are $-3, 3$; 0 has a single preimage, 0; -1 has none in $\mathbb{R}$.

Python

```
def g(x):
    return x*x - 4*x + 32
```

Function, image and preimage

Distinguish `g(-5)` from `x = -5`.

Worked answer: Function, image and preimage

The first calls the function and returns 77. The second assigns -5 to the variable `x`, without calculating the image under g.

Domain of definition

D_f contains the values for which the process defining f makes sense. A denominator must be nonzero; a number under a square root must be positive or zero. Also respect any domain imposed by the statement or the context.

Domain of definition

Determine the domains of definition:

$$f(x) = 3x^2 - 5x + 2, \quad g(x) = \frac{2x+1}{x-4}, \quad h(x) = \frac{x}{x^2-9}, \quad k(x) = \sqrt{2x-6}.$$

Worked answer: Domain of definition

$D_f = \mathbb{R}$. $D_g = (-\infty, 4) \cup (4, +\infty)$ since $x - 4 \neq 0$. $D_h = (-\infty, -3) \cup (-3, 3) \cup (3, +\infty)$ since $x^2 - 9 \neq 0$. $D_k = [3, +\infty)$ since $2x - 6 \geq 0$.

Domain of definition

The square root of a negative number does not exist in $\mathbb{R}$; it is not “a negative square root” that is excluded.

Domain of definition

A rectangle of perimeter 20 m has width x m. Express its area and state the domain if both dimensions must be strictly positive.

Worked answer: Domain of definition

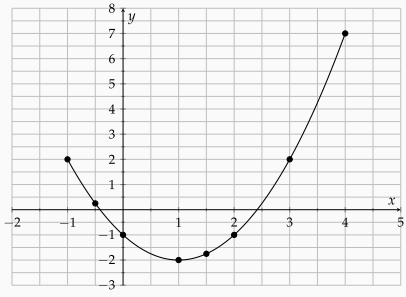
The length is $10 - x$, so $A(x) = x(10 - x)$ in m^2. $x > 0$ and $10 - x > 0$ give $D_A = (0, 10)$. If degenerate rectangles were allowed, we could take $[0, 10]$.

Table of values and representative curve

$\mathcal{C}_f$ is the set of points $(x, f(x))$, with $x \in D_f$. Thus $M(a, b) \in \mathcal{C}_f \iff a \in D_f$ and $b = f(a)$.

Table of values and representative curve

On $[-1, 4]$, $f(x) = x^2 - 2x - 1$. Complete the table, then draw the curve over the whole interval.



Worked answer: Table of values and representative curve

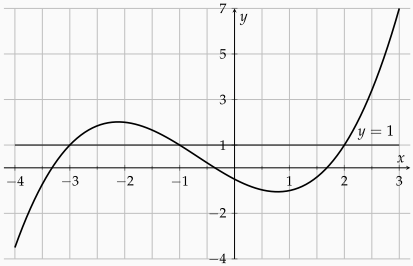
Images: 2; 0.25; -1; -2; -1.75; -1; 2; 7. Plot the points then draw the parabola. Segments joining only these points would give just an approximation.

Graphical solving of equations and inequalities

The solutions of $f(x) = k$ are the abscissae of the intersections of $\mathcal{C}_f$ with $y = k$. For $f(x) \leq k$, read the abscissae where the curve is below this line, intersections included; with $<$, the intersections are excluded.

Graphical solving of equations and inequalities

The curve represents $f(x) = 1 + \frac{1}{4}(x+3)(x+1)(x-2)$ on $[-4, 3]$. Solve graphically $f(x) = 1$, $f(x) \leq 1$ and $f(x) > 1$.



Worked answer: Graphical solving of equations and inequalities

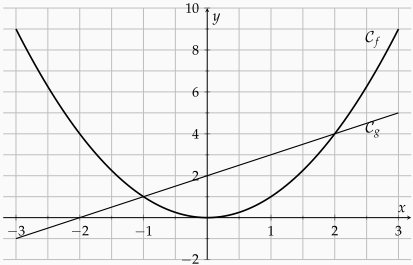
$f(x) = 1$: $-3, -1, 2$. $f(x) \leq 1$: $[-4, -3] \cup [-1, 2]$. $f(x) > 1$: $(-3, -1) \cup (2, 3]$. The endpoints $-4, 3$ come from the domain. The factorisation checks the intersections exactly; the algebraic study of the sign will be consolidated in sequence 8.

Comparing two functions; reading a sign

$f(x) = g(x)$ is read at the intersections of the curves. For $f(x) < g(x)$, $\mathcal{C}_f$ is strictly below $\mathcal{C}_g$ at the same abscissa. The sign of $f(x)$ is read relative to the x-axis.

Comparing two functions; reading a sign

On $[-3,3]$, $f(x) = x^2$ and $g(x) = x + 2$. Solve $f = g$, $f < g$, $f \geq g$ and determine the sign of g.



Worked answer: Comparing two functions; reading a sign

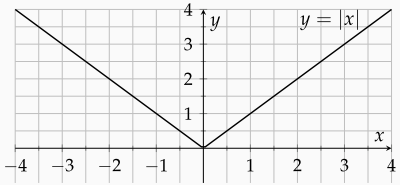
$f = g$ for $x = -1, 2$; $f < g$ on $(-1, 2)$; $f \geq g$ on $[-3, -1] \cup [2, 3]$. g is strictly negative on $[-3, -2)$, zero at -2, strictly positive on $(-2, 3]$. Sign and variations are different pieces of information.

Affine functions and absolute value

An affine function is $f(x) = mx + p$ on $\mathbb{R}$. It is linear if $p = 0$, constant if $m = 0$. For $m \neq 0$, it vanishes at $-p/m$; its sign is that of m to the right of this number. The function $x \mapsto |x|$ is defined on $\mathbb{R}$, positive or zero, and vanishes only at 0.

Affine functions and absolute value

Classify (several answers possible) as affine, linear, constant: $5x + 2$; $5x$; $7 + 2x - 7$; 6; $6(4x - 2)$; $6x + 5 - 6x$.



Worked answer: Affine functions and absolute value

All are affine. $5x$ and $2x$ are linear; 6 and 5 are constant; $24x - 12$ is neither linear nor constant.

Affine functions and absolute value

If $k < 0$, $|x| = k$ and $|x| \leq k$ have no solution, and $|x| \geq k$ is true on $\mathbb{R}$. If $k = 0$, the first two give $x = 0$, the third is true on $\mathbb{R}$. For $k > 0$:

$$|x| = k \iff x \in \{-k, k\}, \quad |x| \leq k \iff x \in [-k, k],$$

$$|x| \geq k \iff x \in (-\infty, -k] \cup [k, +\infty).$$

Affine functions and absolute value

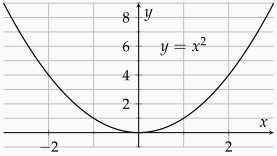
Solve $|x| = 5$, $|x| = -3$, $|x| \leq 3$, $|x| \geq 4$.

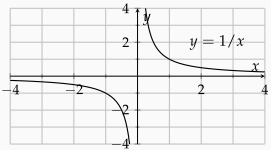
Worked answer: Affine functions and absolute value

$\{-5, 5\}$; $\varnothing$; $[-3, 3]$; $(-\infty, -4] \cup [4, +\infty)$.

Square and inverse functions

$x \mapsto x^2$ is defined on $\mathbb{R}$; its curve is a parabola. $x^2 \geq 0$, with equality only at 0. $x \mapsto 1/x$ is defined on $\mathbb{R}^*$; its curve is a hyperbola. $1/x$ has the sign of x and never vanishes.





Square and inverse functions

If $k < 0$, $x^2 = k$ and $x^2 \leq k$ have no solution; $x^2 \geq k$ is true on $\mathbb{R}$. If $k = 0$, the first two give $x = 0$, the third is true on $\mathbb{R}$. For $k > 0$:

$$x^2 = k \iff x \in \{-\sqrt{k}, \sqrt{k}\}, \quad x^2 \leq k \iff x \in [-\sqrt{k}, \sqrt{k}],$$

$$x^2 \geq k \iff x \in (-\infty, -\sqrt{k}] \cup [\sqrt{k}, +\infty).$$

Square and inverse functions

Solve $x^2 = 4$, $x^2 = -1$, $x^2 \geq 4$, $x^2 \geq -1$, $1/x = 4$, $1/x = 0$.

Worked answer: Square and inverse functions

$\{-2, 2\}$; $\varnothing$; $(-\infty, -2] \cup [2, +\infty)$; $\mathbb{R}$; $\{1/4\}$; $\varnothing$. More generally, $1/x = k$ has the unique solution $1/k$ if $k \neq 0$, none if $k = 0$.

Inequalities with the inverse function

Respect $x \neq 0$ and read both branches of the hyperbola. Do not multiply an inequality by x without knowing its sign.

Inequalities with the inverse function

Solve $1/x \geq 2$, $1/x \leq -1$, $1/x < 1/2$ and $1/x > -2$.

Worked answer: Inequalities with the inverse function

$(0, 1/2]$; $[-1, 0)$; $(-\infty, 0) \cup (2, +\infty)$; $(-\infty, -1/2) \cup (0, +\infty)$. A strict inequality excludes the intersection with $y = k$; 0 is always forbidden.

Inequalities with the inverse function

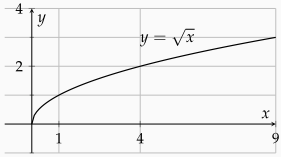
A student writes: "$1/x \leq 2 \iff 1 \leq 2x$, so $x \geq 1/2$". What is missing?

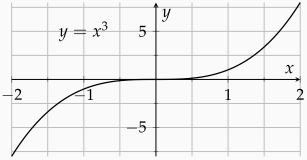
Worked answer: Inequalities with the inverse function

They implicitly assumed $x > 0$. For $x < 0$, $1/x < 0 < 2$: these values also work. The correct set is $(-\infty, 0) \cup [1/2, +\infty)$. The number -1 refutes their equivalence.

Using square root and cube

$x \mapsto \sqrt{x}$ is defined on $[0, +\infty)$ and takes values that are positive or zero. For $k \geq 0$, $\sqrt{x} = k \iff x = k^2$; for $k < 0$, no solution. $x \mapsto x^3$ is defined on $\mathbb{R}$. Every real number k has a unique preimage, written $\sqrt[3]{k}$.





Using square root and cube

Solve $\sqrt{x} = 3$; $\sqrt{x} = -2$; $\sqrt{x} \leq 2$; $\sqrt{x} > 3$; $x^3 = -8$; $x^3 \leq 27$.

Worked answer: Using square root and cube

$\{9\}$; $\varnothing$; $[0, 4]$; $(9, +\infty)$; $\{-2\}$; $(-\infty, 3]$. Respect the domain for square roots. The cube function preserves order.

Using square root and cube

Compare x^2 and x for $x = 1/2$, then $x = 2$. Is the square of a positive number always greater than the number itself?

Worked answer: Using square root and cube

$1/4 < 1/2$ but $4 > 2$. The claim is false; $1/2$ is a counterexample. General study in sequence 10.